Simulating Mechanical Energy Transfer in Rowing: From Thrust–Drag Balance to Disturbance-Sensitive Efficiency with Multibody Dynamics and IMU-Informed Identification

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Abstract. Rowing measurement culture is exceptionally good at quantifying power, yet elite performance depends at least as strongly on the cost of power: losses induced by within-stroke velocity fluctuations, recovery-phase mass motion, and hull attitude dynamics that augment drag and reduce how much propulsive work becomes uninterrupted run. We present a unified modeling framework that (i) reproduces a baseline thrust–drag model with explicit energy accounting, (ii) extends it with reduced-order multibody dynamics capturing rower–shell momentum exchange and pitch-coupled drag augmentation, and (iii) outlines an IMU-informed pipeline for stroke segmentation, disturbance feature extraction, and grey-box parameter identification. The combined study reframes performance optimization as loss minimization: reducing disturbance-induced drag and recovery "check" can yield meaningful speed gains without increasing peak power.

Keywords: rowing; hydrodynamic drag; velocity fluctuations; multibody dynamics; inertial measurement units; system identification; efficiency

INTRODUCTION

Rowing performance emerges from coupled athlete biomechanics, boat dynamics, and fluid mechanics. Even at fixed average power, technique-dependent disturbances—notably recovery-induced center-of-mass (COM) motion and hull attitude oscillations—can increase drag and amplify velocity fluctuations, raising the mechanical work required to maintain race speed. [1, 2, 4] Recent on-water sensing (IMUs and high-rate GPS) enables measurement of acceleration and attitude signatures that correlate with performance and reveal "sub-visible" inefficiencies. [6]

This work contributes: (i) a baseline thrust–drag energy balance model with explicit work and dissipation accounting; (ii) a reduced-order multibody extension capturing rower–shell momentum exchange and pitch-coupled drag augmentation; and (iii) an IMU-to-parameter pipeline for disturbance feature extraction and grey-box identification.

MODELS

Baseline thrust–drag dynamics

We model shell surge velocity $v(t)$ with a periodic drive/recovery forcing:

$$m\,\dot{v}(t) = F_{\text{drive}}(t) - k_d\,v(t)|v(t)| - F_{\text{dist}}(t), \tag{1}$$

where m is the effective system mass and k_d is a quadratic drag coefficient. [2] The drive force $F_{\text{drive}}(t)$ is represented by a smooth pulse over the drive fraction of the stroke; $F_{\text{dist}}(t)$ collects additional losses (e.g., unsteady/attitude drag augmentation or recovery "check") introduced in later sections.

Energy bookkeeping follows from multiplying Eq. (1) by v and integrating over a stroke period T: input work $W_{\text{in}} = \int_0^T F_{\text{drive}} v\,dt$, drag dissipation $E_D = \int_0^T k_d|v|^3 dt$, and net kinetic change $\Delta K = \frac{1}{2}m(v^2(T) - v^2(0))$.

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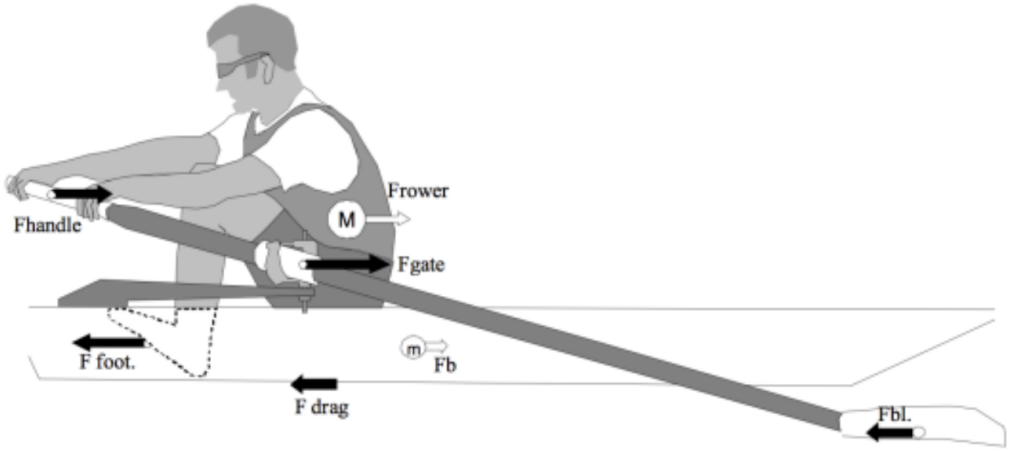


FIGURE 1. Free-body schematic of the rowing shell surge model. Propulsive drive forces transmitted through the oar/rigging oppose hydrodynamic resistance; technique-dependent disturbances are represented as an additional loss term and (in the extended model) as pitch-coupled drag augmentation.

Multibody COM exchange and pitch-coupled drag

To expose technique-dependent loss channels, we separate rower COM motion from shell motion using a reduced-order two-mass model (rower on slide + shell) with internal force exchange. [1, 3] A pitch/trim surrogate $\theta(t)$ is coupled to recovery-phase mass motion and to drag via a small-angle augmentation term, motivating a disturbance-sensitive effective drag $k_d \to k_d(1+\alpha_\theta\theta^2+\alpha_{\dot\theta}\dot\theta^2)$. [5]

IMU-informed disturbance features and identification

Given 6-axis IMU measurements, we estimate orientation, remove gravity, segment strokes, and compute time–frequency features (e.g., wavelets or short-time spectra) that quantify transient disturbances. [6] A grey-box identification step then fits parameters (e.g., k_d, disturbance amplitudes, and pitch-coupling coefficients) by minimizing the discrepancy between simulated and measured acceleration/attitude signatures.

RESULTS

Figure 1 summarizes the force pathways used throughout the model: drive-phase handle force is transmitted through the oar/rigging to produce net surge thrust, opposed by hydrodynamic resistance and augmented losses associated with within-stroke disturbances (e.g., recovery "check" and pitch-coupled drag). In the baseline limit, the model reduces to a periodic thrust–drag balance with transparent energy accounting; supplementary stroke-resolved velocity/acceleration and energy metrics are reported in the Appendix.

Figure 2 shows the velocity-dependent resistance parameterization used for simulations reported here. A fitted $C_D(v)$ (or equivalently an effective $k_d(v)$) supports stroke-resolved dissipation estimates while accommodating regime changes in hull resistance.

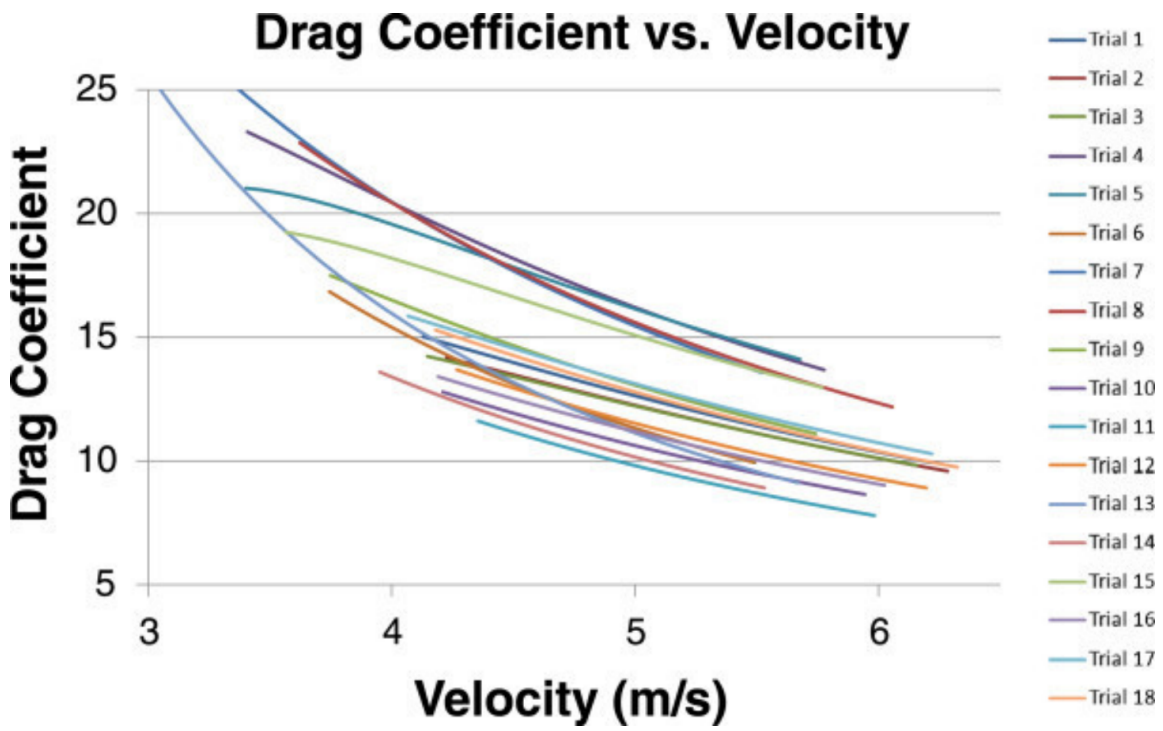


FIGURE 2. Velocity-dependent resistance fit used in the simulations. The fitted drag coefficient (or effective quadratic drag) is used to compute stroke-resolved dissipation and to compare "smooth" versus disturbance-augmented cases at fixed mean power.

DISCUSSION

The unified framework emphasizes that performance is not determined solely by producing power, but by minimizing the mechanisms that *waste* that power: velocity oscillation penalties, recovery-induced "check," and attitude-driven drag augmentation. [4] IMU-informed identification provides a path to estimate these penalties on-water and to connect measurable signatures (acceleration, pitch) to interpretable parameters.

CONCLUSION

A periodic thrust–drag baseline reproduces within-stroke dynamics and enables transparent energy accounting. Extending to multibody COM exchange and pitch-coupled drag makes technique-dependent losses explicit. Combined with an IMU pipeline and grey-box identification, the approach supports a data-driven pathway for measuring and minimizing technique-driven losses.

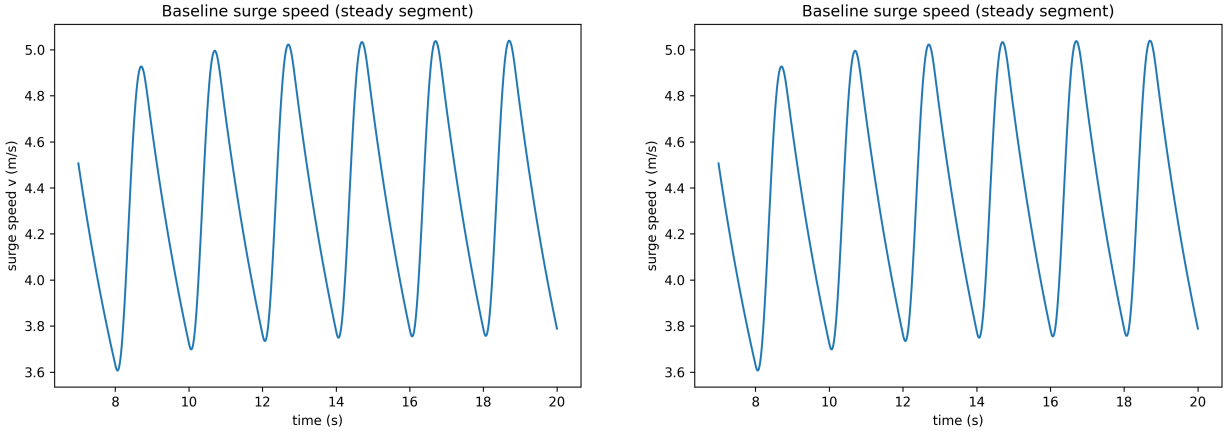
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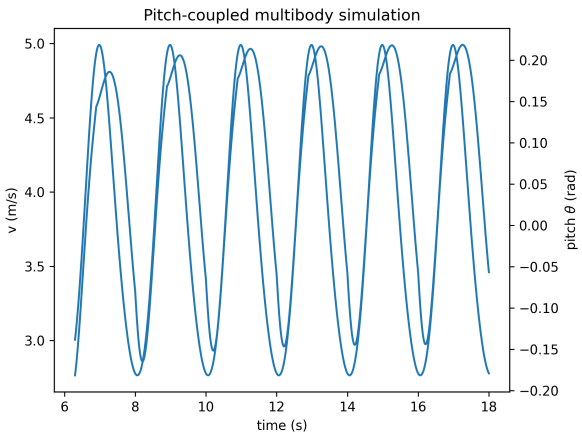
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Supplementary Material

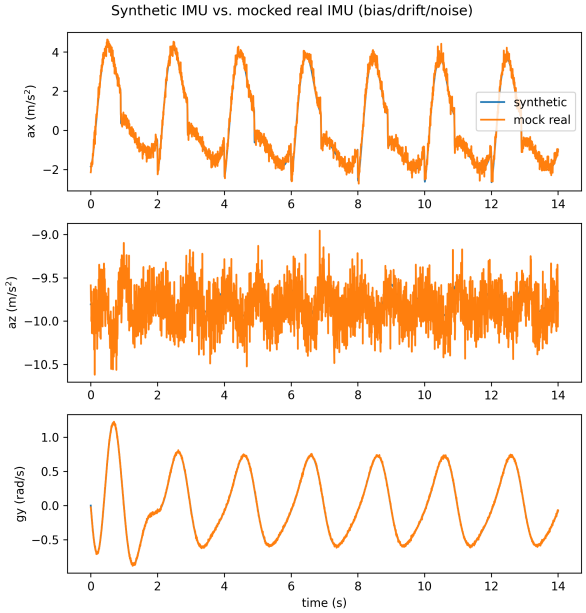


(a) (b)

FIGURE 3. Baseline thrust–drag outputs. (a) Stroke-resolved surge velocity/acceleration in a steady segment. (b) Work and dissipation bookkeeping used to compute efficiency and to isolate the cost of within-stroke velocity oscillations.



(a)



(b)

FIGURE 4. Extended reduced-order and sensing outputs. (a) Example pitch-coupled drag surrogate showing how attitude dynamics can modulate resistance over the stroke. (b) Illustrative IMU workflow: orientation estimation, gravity removal, stroke segmentation, and time–frequency disturbance features.

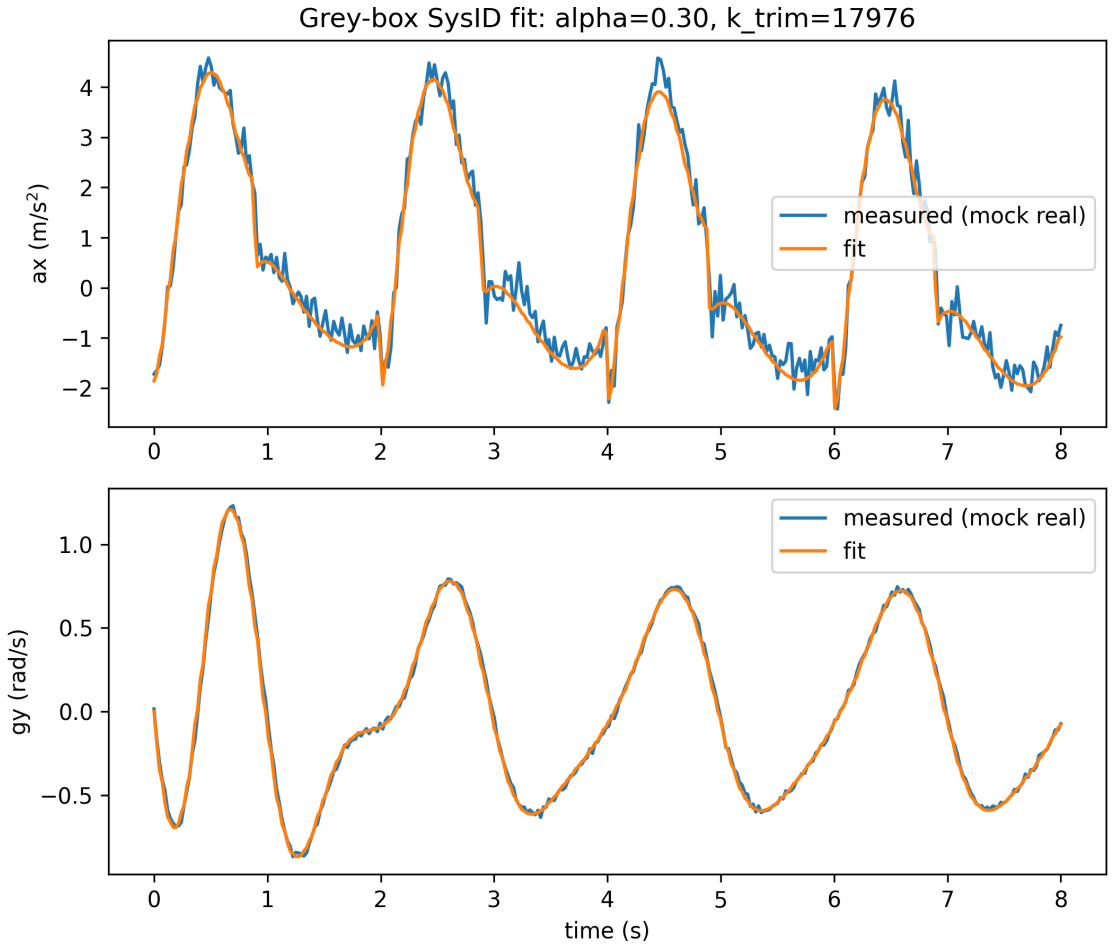


FIGURE 5. Example grey-box identification output supporting parameter estimation (e.g., k_d, disturbance amplitudes, and pitch-coupling coefficients) from simulated or measured IMU signatures.